

Digital Design SET 111

Lecture (03)- Number Systems, Operations, and Codes

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- ☐ **Decimal Number**
- ☐ **Binary Number**
- ☐ **Decimal-to-Binary Conversion**
- ☐ **Binary Arithmetic**
- ☐ **Complements of Binary Numbers**



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Decimal Number



- ❑ In the decimal number system, each of the **ten** digits, $0 \rightarrow 9$, represents a certain quantity.
- ❑ The position of each digit in a **decimal number** has its own represented **weight**.
- ❑ The value of a decimal number is the sum of the digits after each digit has been multiplied by its weight.
- ❑ **Ex:** Express the decimal number $(568.23)_{10}$ as a sum of the values of each digit

Weight	10^2	10^1	10^0		10^{-1}	10^{-2}	
	↓	↓	↓		↓	↓	
Decimal number	5	6	8	.	2	3	
The value	$5 \cdot 10^2$	$+$	$6 \cdot 10^1$	$+$	$8 \cdot 10^0$	.	$2 \cdot 10^{-1} + 3 \cdot 10^{-2}$
	500	$+$	60	$+$	8	.	0.2 + 0.03 = $(568.23)_{10}$
							Decimal number



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Binary Number (1/3)

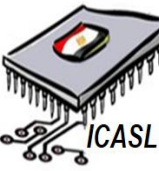


- ❑ The **binary** number system is another way to represent quantities
- ❑ The two binary digits (bits) are **1** and **0**.
- ❑ **Ex. Counting in Binary**
 - ❑ **B0** → Least Significant Bit (**LSB**)
 - ❑ **B2** → Most Significant Bit (**MSB**)
 - ❑ Largest decimal number = $2^n - 1$
 - ❑ n : represents the number of bits
 - ❑ Ex, 3 bits ($n = 3$) you can count from 0 to 7

Ex, 2 bits ($n = 2$) you can count from 0 to 3

Decimal Number	Binary Number		
	B2	B1	B0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

Binary Number (2/3)



- ❑ In the binary number system, each of the **binary** digits {1,0}, represents a certain quantity.
- ❑ The position of each digit in a **binary number** has its own represented **weight**.
- ❑ The **binary number** can be **converted** into **decimal number** by summing of the digits after each digit has been multiplied by its weight.
- ❑ Ex: Convert the binary whole number $(1101101.1011)_2$ to decimal.

Weight	2^6	2^5	2^4	2^3	2^2	2^1	2^0	.	2^{-1}	2^{-2}	2^{-3}	2^{-4}
	↓	↓	↓	↓	↓	↓	↓		↓	↓	↓	↓
Binary number	1	1	0	1	1	0	1	.	1	0	1	1
The value of binary number in decimal	64	+ 32	+ 0	+ 8	+ 4	+ 0	+ 1	+	0.5	+ 0	+ 0.125	+ 0.0625 = (109.6875) ₁₀

$$(1101101.1011)_2 = (109.6875)_{10}$$

Binary-to-Decimal Conversion

Binary Number (3/3)



❑ Application

❑ Simple binary counting system.

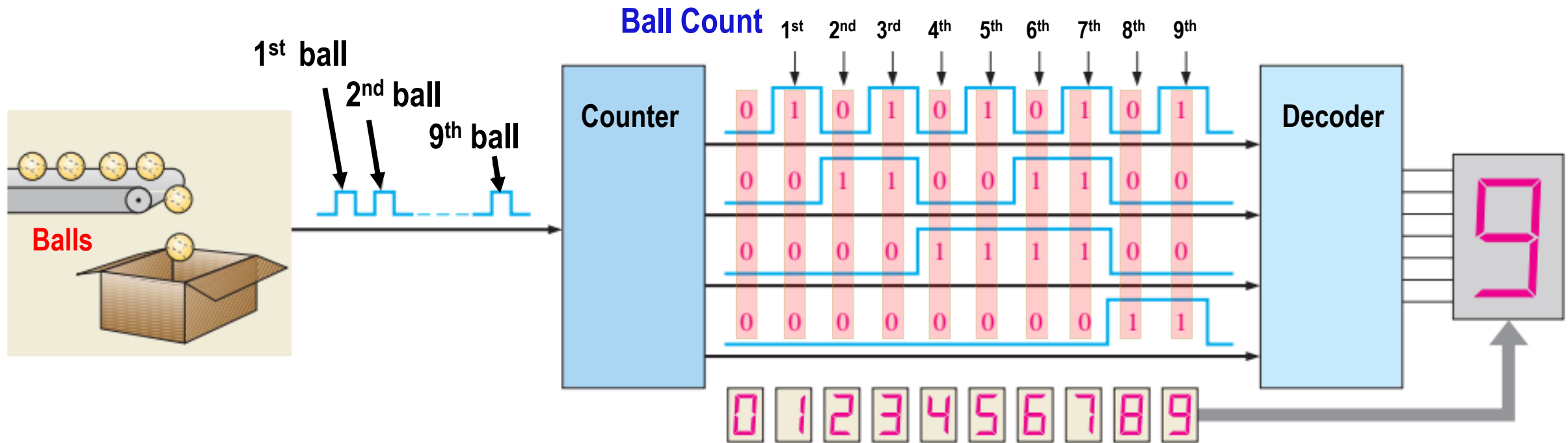


Illustration of a simple binary counting application



- ☐ Decimal Number
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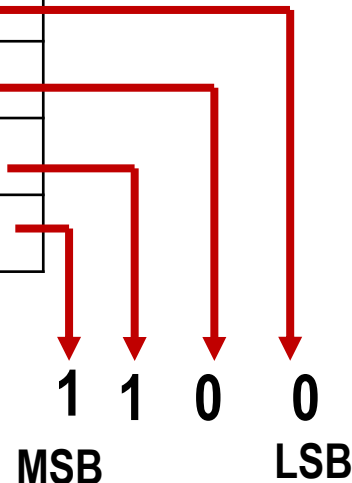
Decimal-to-Binary Conversion (1/2)



- ❑ Decimal number could have **whole** and **fractional** $\rightarrow (45.3125)_{10}$
- ❑ A systematic method of converting **whole** numbers from decimal to binary is the repeated **division-by-2** process until the **quotient** is **0**.
- ❑ The generated **remainders** by each division **form** the **binary** number
 - ❑ Ex.1.- Convert the decimal number $(12)_{10}$ to binary
 - ❑ Ex.2.- Convert the $(45)_{10}$ decimal numbers to binary

Step	quotient	Remainder
12/2	6	0
6/2	3	0
3/2	1	1
1/2	0	1

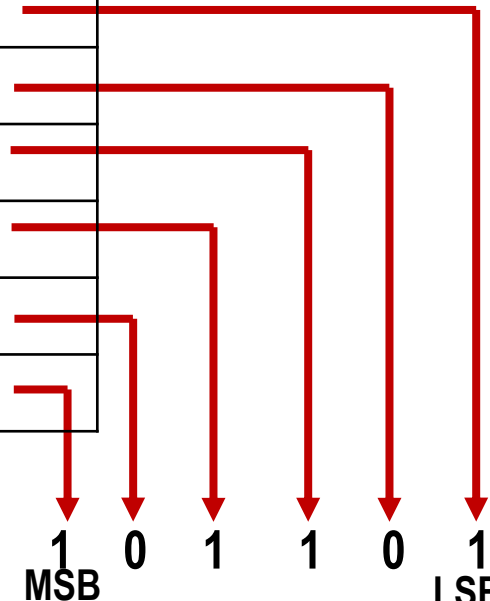
Stop when the Whole-number quotient is 0



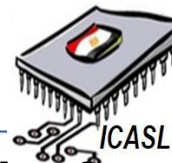
$$(12)_{10} = (1100)_2$$

Step	quotient	Remainder
45/2	22	1
22/2	11	0
11/2	5	1
5/2	2	1
2/2	1	0
1/2	0	1

Stop when the Whole-number quotient is 0



$$(45)_{10} = (101101)_2$$



Decimal-to-Binary Conversion (2/2)

- ❑ A systematic method of converting **Decimal Fractions** to **Binary** is the **repeated Multiplication by 2** process until the **fractional product is zero** **or** until the **desired number of decimal places is reached** .
- ❑ The generated carry digits, or **carries**, by the **multiplications** produce the **binary** number
- ❑ EX: convert the decimal number $(0.3125)_{10}$ to binary

Step	Fractional	Carry	MSB			LSB
			0	1	0	1
0.3125×2	0.625	0				
0.625×2	$1.25 \rightarrow 0.25$	1				
0.25×2	0.5	0				
0.5×2	$1.0 \rightarrow 0$	1				

Continue to the desired number of decimal places or stop when the fractional part is all zeros

$(0.3125)_{10} = (0.0101)_2$

$(45.3125)_{10}$

Step1: Convert the Whole Part
 $(45)_{10} = (101101)_2$

Step:2 Convert the Fraction Part
 $(0.3125)_{10} = (0.0101)_2$

$(45.3125)_{10} = (101101.0101)_2$



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- ☐ Decimal-to-Binary Conversion
- ☐ **Binary Arithmetic**
- ☐ Complements of Binary Numbers

Binary Arithmetic (1/2)



❑ Binary arithmetic is essential in all digital computers and in many other types of digital systems

❑ Binary Addition

❑ The **four** basic rules for adding binary digits (bits)

❑ Ex:- Add the binary numbers $(11)_2$ to $(01)_2$

❑ Ex:- Add the binary numbers $(111)_2$ to $(11)_2$

A	B	Sum	Carry out
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Carry 1 1 1

A	1	1	1	
B	1	1	1	+
Result	1	0	1	0

Carry 1 1

A	1	1	
B	0	1	+
Result	1	0	0

Carry in	A	B	Sum	Carry out
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Binary Arithmetic (2/2)



❑ Binary arithmetic is essential in all digital computers and in many other types of digital systems

❑ Binary Subtraction

❑ The **four** basic rules for subtracting binary digits (bits)

	A	B	Subtract
	0	0	0
1	0	1	1
	1	0	1
	1	1	0

❑ Special Case:

❑ A **borrow** is required in binary only when you try to subtract a 1 from a 0

❑ Ex:- Subtract $(011)_2$ from $(101)_2$

Right column:

$$1-1=0$$

Middle Column:

Borrow 1 from next column to the left, making a 10 in this column, then $10 - 1 = 1$

Left column:

When a 1 is borrowed a 0 is left, so $0 - 0 = 0$

$$\begin{array}{r}
 0 \\
 A 1^1 0 1 \\
 B 0 1 1 \\
 \hline
 0 1 0
 \end{array}$$

$$\begin{array}{r}
 0 \\
 A 1^1 0 \\
 B 1 \\
 \hline
 1
 \end{array}$$



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Complements of Binary Numbers (1/3)



❑ The **1's complement** and the **2's complement** of a binary number are important because they permit the representation of **negative** numbers.

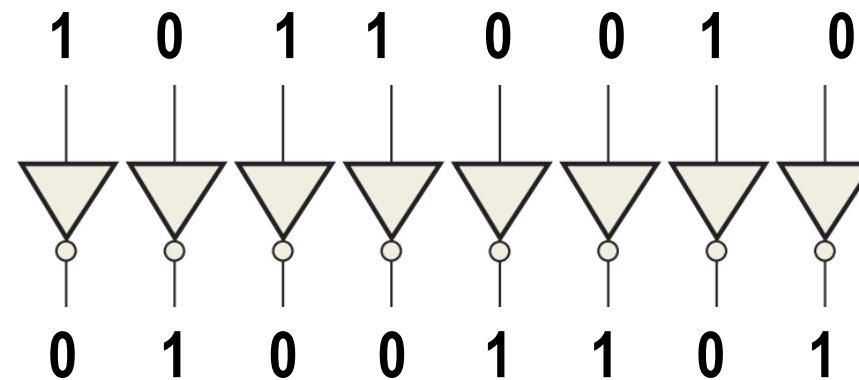
❑ 1's complement

❑ is obtained by changing each **1** $\rightarrow$ **0** and each **0** to **1**.

❑ Ex:- Find the 1's complement of $(10110010)_2$.

❑ In **hardware**, array of inverters can be used to provide the **1's complement**

Binary Number	1	0	1	1	0	0	1	0
	↓	↓	↓	↓	↓	↓	↓	↓
1's complement	0	1	0	0	1	1	0	1



Hardware Digital circuit for 1's complement

Complements of Binary Numbers (2/3)



□ 2's complement

□ 1st method:

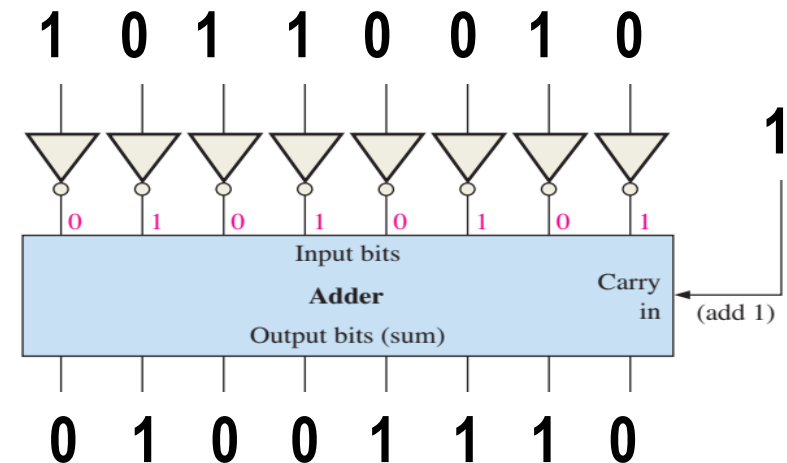
□ 2's complement of a binary number is found by adding 1 to the LSB of the 1's complement

$$\text{2's complement} = (\text{1's complement}) + 1$$

□ Ex.1 . Find the 2's complement of $(10110010)_2$, using the 1st method.

□ In **hardware**, array of inverters can be used with an adder

Binary Number	1	0	1	1	0	0	1	0
	↓	↓	↓	↓	↓	↓	↓	↓
1's complement	0	1	0	0	1	1	0	1
								+
								1
2's complement	0	1	0	0	1	1	1	0



Hardware Digital circuit for 1's complement

Complements of Binary Numbers (3/3)

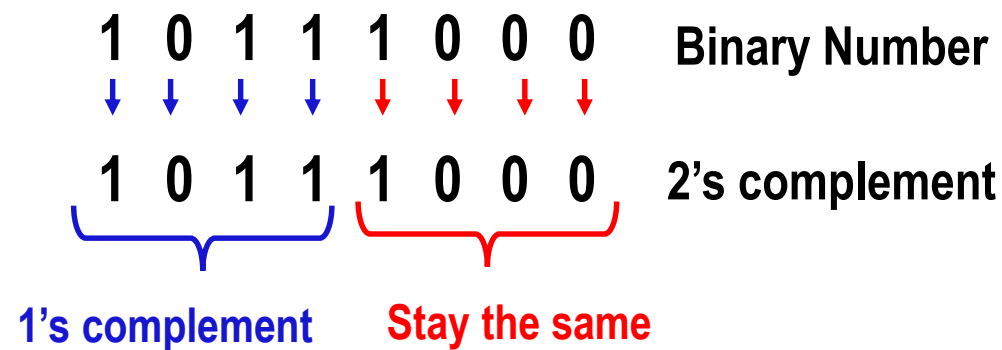


□ 2's Complement

□ 2nd method:-

- Start at the right with the LSB and write the bits as they are up to and including the 1st 1.
- Take the 1's complements of the **remaining bits**.

□ Ex.1 Find the 2's complement of $(10111000)_2$ using the 2nd method



Thanks!